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RESEARCH

Regional scale analysis of the internal propagation of groundwater droughts in Andhra Pradesh, India

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Abstract

Accurate characterization of groundwater droughts is critical for efficiently managing groundwater resources at multiple scales. Over the past years, most of the research has been focused on analyzing propagation characteristics from meteorological droughts to groundwater droughts; however, a clear understanding of the internal propagation of groundwater droughts at the regional scale is lacking. Therefore, in this study, we investigate the internal propagation of groundwater droughts and analyze various drought characteristics at a regional scale in Andhra Pradesh state. From this study, we tried to understand the groundwater drought propagation speeds and the relation between groundwater drought characteristics across three different regions in Andhra Pradesh. Nonparametric Standardized Groundwater Index (SGI) and Groundwater Drought Instantaneous Propagation Speed (GDIPS) are used to study groundwater drought characteristics and internal propagation dynamics. Our results indicate that groundwater drought internal propagation characteristics are highly spatially variable, and even high-rainfall regions are susceptible to prolonged groundwater droughts. Groundwater droughts in the study area are characterized as mild to moderately frequent. Drought Recovery Duration (DRD) influences the cyclic nature of groundwater time series in most districts of the study region. This study helps understand the groundwater drought internal propagation mechanism and effectively manage groundwater resources from overexploitation.

Highlights

- The Groundwater Drought Instantaneous Propagation Speed (GDIPS) framework is developed to characterize the internal propagation of groundwater droughts at the regional scale.
- Contrasting drought development and recovery patterns are observed across the study region.
- Groundwater drought recovery duration is highly correlated with the temporal dynamics of groundwater time series.

1 Introduction

Droughts are natural or human-induced hazards that cause significant loss to human habitats, agriculture, and ecosystem services. Droughts are generally classified into meteorological (precipitation deficit), agricultural (soil moisture deficit), hydrological (low streamflow or water table), and socio-economic droughts (supply deficit to meet demand). Typically, meteorological drought is considered the starting phase of drought in the hydrological cycle and later propagates through various stages of the water cycle (soil moisture deficit and low streamflow or water table). However, the drought propagation mechanism is complex, and meteorological drought may not necessarily be the onset of other droughts (Van Loon 2015; Van Loon et al. 2016). Groundwater drought is a hydrological drought due to sustained low water table and stream discharges. Climate change, land use and land cover changes, population growth, and extensive groundwater irrigation are the driving forces exacerbating groundwater drought conditions. The insufficient recharge for groundwater replenishment further exacerbates the drought conditions (Aeschbach-Hertig et al. 2012). The various hydrological components are nonlinearly correlated, making the drought propagation process more complex. Drought propagation refers to the growth of the deficit and its recovery at the end of a drought

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spell (Bhardwaj et al. 2020). Climate change and human interventions substantially affect the nonlinear process of drought propagation. Hence, it is critical to understand the characteristics and complex propagation mechanisms of groundwater droughts at multiple scales to manage groundwater resources effectively.

Many researchers studied the spatiotemporal variation of groundwater drought characteristics (duration, severity, and intensity) and propagation patterns from meteorological to groundwater droughts. Peters et al. (2006); Tallaksen et al. (2009) assessed the spatiotemporal distribution of groundwater drought characteristics in the Pang catchment in the UK using the constant threshold level method. In another study (Heudorfer and Stahl 2017), the selection of constant and variable threshold level methods was compared to study the spatiotemporal variation of drought propagation in different catchments in Germany. High spatiotemporal variation of groundwater droughts was observed in the South Korean catchment study led by Lee et al. (2018). Wossenyeleh et al. (2021,2022) applied variable threshold methods and numerical groundwater models to identify groundwater and meteorological droughts in central Belgium and Ethiopia. The spatial variability of rainfall significantly influenced the spatial variation of groundwater recharge (Van Lanen and Peters 2002). Most of the studies mentioned above focused on understanding the propagation patterns from meteorological to groundwater droughts; however, a clear understanding of the internal propagation mechanisms of groundwater droughts is lacking. Various techniques employed in the groundwater drought characterization are mentioned in the paragraphs below.

Standardized drought indices have been extensively used to investigate the spatiotemporal variation of groundwater droughts. A study (Fendeková and Fendek 2012) focused on developing a standardized base flow index that compared the base flow severity in the sub-basins. Extremely severe base flow droughts were frequently observed in the mountainous part of the Nitra River basin. A new drought index called the Standardized Groundwater Index (SGI), which uses a nonparametric approach (Bloomfield and Marchant 2013; Bloomfield et al. 2015; Marchant and Bloomfield 2018; Brauns et al. 2020), was used to investigate different groundwater drought characteristics and the propagation of meteorological drought to groundwater drought conditions in the UK. Several studies focused on the propagation of meteorological droughts to groundwater droughts (Liu et al. 2016; Kumar et al. 2016; Leelaruban et al. 2017; Guo et al. 2021; Balacco et al. 2022; Babre et al. 2022). Most studies performed cross-correlation analyses between the Standardized Precipitation Index (SPI) at different accumulation periods and monthly Standardized Groundwater Index (SGI) values in their respective study areas. In most studies, long-duration and less severe groundwater droughts were commonly observed in all the study areas compared to meteorological droughts. Another study (Goodarzi et al. 2016) focused on developing a new groundwater drought index using simulated recharge values in the Najafabad plain in the Ishafahan province of Iran. The authors highlighted that the Groundwater Recharge Drought Index (GRDI) effectively represented climate change and human effects on the spatiotemporal distribution of groundwater droughts. Another study proposed using time series modeling to predict and analyze hydrological data related to extreme conditions (Soleimani et al. 2017). In that study, the main objective was to investigate drought propagation through time series modeling and to analyze the relationship between SPI and SGI in heterogeneous zones of the study area. Other researchers have estimated the frequency of groundwater drought cycles using wavelet transform analyses (Huang et al. 2021).

With the development of remote sensing technologies, like Gravity Recovery and Climate Experiment (GRACE) and GRACE Follow-On missions, groundwater drought research has extended to continental and national scales. Several studies have focused on applying Groundwater Storage (GWS) anomalies derived from GRACE satellite data in conjunction with the standardized drought indices to study the spatiotemporal distribution of groundwater droughts(Thomas et al. 2017; Van Loon et al. 2017; Han et al. 2019; Shah and Mishra 2020; Li and Rodell 2021). However, the coarse spatial resolution of the GRACE and GRACE-FO satellite data restricted their usage at regional scales where more complex propagation patterns may be available.

Most of the work done in the past was focused on studying groundwater drought characteristics through threshold-level methods, standardized drought indices, and remote sensing techniques. The cross-correlation method was mainly used to determine the groundwater's response time to meteorological droughts. Few researchers have explored how groundwater droughts spread internally, highlighting the need to understand these complex processes better. Moreover, understanding groundwater drought internal propagation mechanisms at a state level is lacking, which helps to manage groundwater resources better. Therefore, our main objective is to develop a groundwater drought instantaneous propagation speed (GDIPS) framework to understand the groundwater drought's internal propagation characteristics at the regional scale in Andhra Pradesh. We further examined the spatial variation of groundwater drought characteristics and performed cross-correlation analyses between multiple groundwater drought characteristics and the maximum autocorrelation lag of the groundwater time series. This study also examines the role of rainfall on groundwater

drought dynamics across the study region. This study helps water managers and policymakers better understand the spatial variation of development and recovery phases of groundwater droughts at a state or regional scale.

2 Study area and data

Andhra Pradesh is a highly populated and the seventh-largest state, covering 1,63,000 km^2 in India (Fig. 1). Andhra Pradesh is located between 12° 37′ and 19° 09′ north latitudes and 76° 45′ and 84° 47′ east longitudes and is administered by 13 districts (before 4th April 2022). The study area is divided into three regions: Rayalaseema (RAY), North coastal Andhra (NCA), and South Coastal Andhra (SCA). A gneissic complex with a structural fill of sedimentary formations and a basin fill of meta-sedimentary formations and meta-sediments underlies the central part of the study area. The state's average annual rainfall is 967 mm and has a significant seasonal variation. Southwest monsoon contributes nearly 58% of the average annual rainfall, whereas 30% and 12% occur in the northeast and non-monsoon seasons, respectively (Groundwater yearbook 2020-21 Andhra Pradesh, CGWB). The flow chart illustrating the methodology of the present study is represented in Fig. 2. District-wise monthly water table depth observations and rainfall data from June 2004 to May 2021 were obtained from the India Water Resources Information System (India-WRIS) portal (https://indiawris.gov.in/wris/). District-wise monthly water table depth observations were obtained from shallow monitoring wells that predominantly tap unconfined aquifers across the state. In these aquifers, the water table responds directly to seasonal recharge, evapotranspiration, and pumping stresses, making them suitable for analyzing the internal propagation of groundwater drought conditions. A total of 1252 groundwater monitoring wells were found distributed throughout the state. Out of these 1252 monitoring wells, wells with more than 95% (i.e., > 194 months) of data, were initially screened for the analysis. Hence, 420 out of 1252 monitoring wells (nearly 34%) were analyzed in the present study. Linear interpolation is used to impute any small gaps available in the groundwater data.

3 Methodology

3.1 Modified Mann-Kendall trend test

The Mann-Kendall trend test (MK test) is a nonparametric test used to identify the presence of a monotonic increase or decrease in the data. A monotonic increase or decrease in the water table levels significantly affects groundwater drought development (or recovery) and propagation mechanisms. Many studies employed trend analysis to identify significant trends in groundwater time series. Jackson et al. (2015) analyzed 14 bore wells in the Chalk aquifer in the UK to

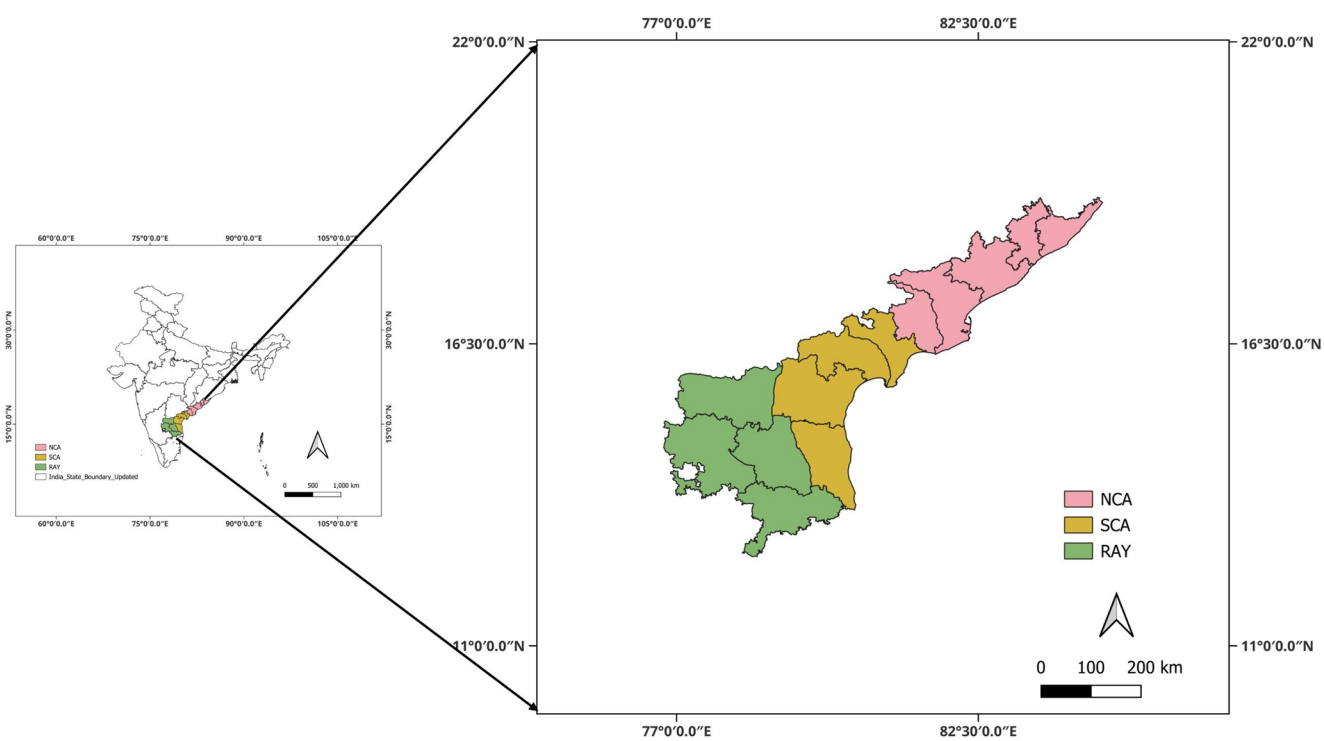

Fig. 1 Geographical representation of Andhra Pradesh state with different regions

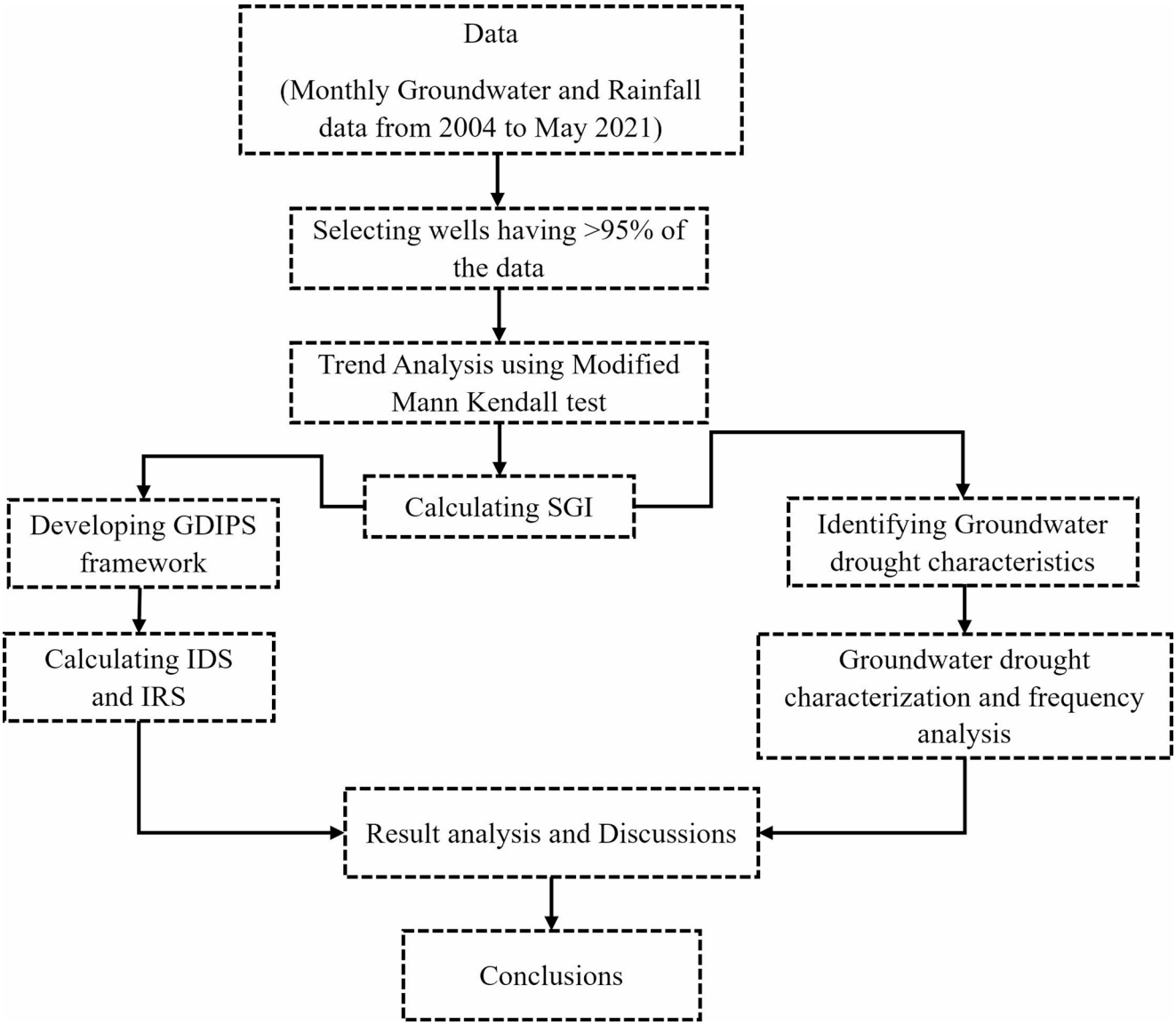

Fig. 2 Flow chart depicting the methodology of the present study

detect any trend due to climate change. 13 wells out of 14 showed a significantly decreasing trend in the standardized groundwater index values. Other researchers have also used the Mann-Kendall test to identify a significant trend in groundwater levels under extreme climate cycles (Le et al. 2018; Haas and Birk 2019). However, serial correlation in the data distorts the trend estimation as hydrological data, like water table depths, are expected to have a significant autocorrelation component. Hence, estimating trends considering the inherent serial correlation of groundwater data is required to identify the anthropogenic influence on groundwater time series. A modified nonparametric Mann-Kendall test was proposed based on the modified value of the variance of the original Mann-Kendall trend test statistics (Hamed and Ramachandra Rao 1998). The modified Mann-Kendall trend test employs the subtraction of a nonparametric trend estimate from the time series and estimates the autocorrelation between the conventional ranks of the observations. The modified MK test outperformed the original MK test without losing power in estimating the presence of a significant trend. We perform trend analysis for all 420 groundwater monitoring wells in the study area using the "*PyMannKendall*" Python package (Hussain and Mahmud 2019). Hamed and Rao's Modified MK test function is used in the Python package. Of the 420 groundwater monitoring wells, 114 have no significant trend (increasing or decreasing) at a 5% significance level (α=0.05). Therefore, only these 114 wells distributed throughout the 13 districts of the study area were chosen to analyze groundwater drought internal propagation unaffected by anthropogenic activities.

3.2 Standardized groundwater index (SGI)

Several studies employed standardized index methods to investigate the spatiotemporal distribution of groundwater drought characteristics and propagation patterns. The application of standardized drought indices in drought research became popular due to the possibility of comparing various drought characteristics under multiple scales. We use a nonparametric standardized groundwater index (SGI) based on a normal score transform to investigate the occurrence of groundwater droughts in the study area. Bloomfield and Marchant (2013) applied the normal score transform, a nonparametric normalization technique, to monthly groundwater levels that assigns a value to observations based on their rank in the data set. The normal score transform is applied by applying the inverse normal cumulative distribution function to n equally spaced p_i (probability) values ranging from (1/2n) to 1–1/(2n), and that results in the SGI values (Eqs. 1–3). The SGI values are reordered such that the largest SGI value is assigned to the *i* for which p_i is the largest, and the second-largest SGI value is assigned to the *i* for which p_i is the second largest, and so on. Many studies (Kumar et al. 2016; Guo et al. 2021) assumed groundwater levels and precipitation data follow various probability distribution functions (normal, gamma, log-normal, extreme value, etc.). However, nonparametric SGI was developed considering the difficulty of fitting one probability distribution to all the groundwater monitoring wells or large data sets. This motivated us to use the nonparametric SGI in the present study. "*Pastas*", a Python package (Collenteur et al. 2020), is used to compute SGI values for the groundwater data in the study area. Significant spatiotemporal variations of groundwater droughts are obtained in the study area (Fig. S1). Because the monitoring wells predominantly represent unconfined aquifers, the SGI derived from water table fluctuations reflects near-surface storage dynamics rather than pressure changes in confined systems.

$$x_1 < x_2 < x_3 \cdots < x_n$$

$$p_i = \frac{i - 0.5}{n},\ i = 1,2,3 \cdots, n \tag{1}$$

$$s_i = \varnothing^{-1}(p_i) \tag{2}$$

$$SGI_t = s_{rank}(x_t) \tag{3}$$

Where x_t represents the monthly water level observations in the monitoring wells, p_i is the empirical cumulative probability, $\varnothing^{-1}$ is the inverse standard normal cumulative distribution function, and s_i are the corresponding standard normal scores. The SGI time series represents standardized anomalies of monthly water table fluctuations, with negative values representing groundwater droughts, whereas the positive values indicate higher-than-normal values.

3.3 Groundwater drought instantaneous propagation speed (GDIPS)

Substantial research was focused on drought propagation from meteorological to groundwater droughts (Han et al. 2019; Soleimani et al. 2017); however, studies on internal propagation of groundwater droughts are limited (Parry et al. 2018), and the spatiotemporal variation of groundwater drought under development and recovery phases remains unclear. Internal propagation of drought occurs between the development and recovery phases (Bonsal et al. 2011). Wu et al. (2018) and Bhardwaj et al. (2020) used Hydrological Drought Instantaneous Propagation Speed (HDIPS) based on the variable motion relationship of the speed-time process to study the variation of streamflow drought during the development and recovery phases. Parry et al. (2018) investigated the UK's internal groundwater drought termination behavior but considered a constant recovery rate throughout the recovery period. The limitation of a uniform recovery rate during the recovery phase was addressed by Wu et al. (2018), considering non-uniformity in the streamflow drought development and recovery phases. Groundwater droughts are also expected to have a significant non-uniform internal drought propagation mechanism. Hence, we followed the HDIPS framework to develop Groundwater Drought Instantaneous Propagation Speed (GDIPS), which determines the speed-time process relationship of groundwater drought propagation. According to our knowledge, no study has adopted this technique to study the internal propagation of groundwater droughts in Andhra Pradesh. We identified groundwater drought events for each well using the run theory (Yevjevich, 1967) (Fig. 3). According to the run theory, if the drought index is negative continuously for

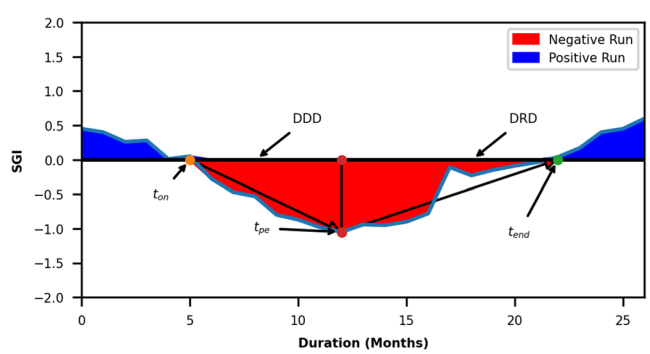

Fig. 3 Schematic representation of run theory used to extract DDD (or DRD). DDD is the duration between t_{pe} and t_{on}, and DRD is between t_{end} and t_{pe}

a threshold period of time, then the run is considered a negative run; if the drought index becomes positive continuously for a specific threshold period of time, it is a positive run. In this study, groundwater drought onset is defined when monthly SGI values are less than zero continuously for at least three months (negative run), and drought termination is considered when monthly SGI values are greater than zero continuously for at least three months (positive run).

After identifying various groundwater drought episodes for each monitoring well, we extracted Drought Development Duration (DDD, the duration between the onset period (t_{on}) to the peak drought period (t_{pe}) (minimum 1-month SGI value)), and Drought Recovery Duration (DRD, duration between peak drought period (t_{pe}) to drought end period (t_{end})). The Total Variation in the Development Period (TVDP) and Total Variation in the Recovery Period (TVRP) are determined by calculating the cumulative sum of variation of monthly SGI values during drought development and recovery stages as per Eqs. 6 and 7. During the TVDP (or TVRP) calculation, the monthly SGI value is expected to reach a minimum value at t_{pe} and a positive or zero value after t_{end}. Hence, all DDD and DRDs are divided into *n* time intervals, and $\Delta TVDP_i$ and $\Delta TVRP_i$ are calculated for each time interval as per Eqs. 4 and 5. The positive values of $\Delta TVDP_i$ and negative values of $\Delta TVRP_i$ are neglected to eliminate temporary recovery and development in drought development and recovery phases, respectively. Moreover, during long-duration drought events, multiple peak intensity values are common; the minimum value among the multiple peak intensity values is considered the threshold between the development and recovery phases. Using the TVDP (or TVRP) values, Instantaneous Development Speed (IDS) and Instantaneous Recovery Speed (IRS) are calculated, which give information on how fast the development or recovery of a drought event is taking place. IDS (or IRS) will have the units equal to the change in drought index per month; in the present study, ΔSGI/month or ΔSGI.month^{-1}.

$$\Delta\,\mathrm{TVDP_i} = (\mathrm{SGI_{i+1}} - \mathrm{SGI_i}) \quad \text{for i} = 1,2,3\ldots..\mathrm{n} \tag{4}$$

$$\Delta\,\mathrm{TVRP_i} = (\mathrm{SGI_{i+1}} - \mathrm{SGI_i}) \quad \text{for i} = 1,2,3\ldots..\mathrm{n} \tag{5}$$

$$\mathrm{TVDP} = \sum{}_{\mathrm{i=1}}^{\mathrm{n}}\,(\Delta\,\mathrm{TVDP_i}) \quad \text{for i} = 1,2,3\ldots..\mathrm{n} \tag{6}$$

$$\mathrm{TVRP} = \sum{}_{\mathrm{i=1}}^{\mathrm{n}}\,(\Delta\,\mathrm{TVRP_i}) \quad \text{for i} = 1,2,3\ldots..\mathrm{n} \tag{7}$$

Average Instantaneous Development Speed ($\overline{\mathrm{IDS}}$) and Average Instantaneous Recovery Speed (the $\overline{\mathrm{IRS}}$) are calculated as per Eqs. 8 and 9. In the present study, $\Delta\,\mathrm{T}$ is equal to 1 month.

$$\overline{\mathrm{IDS}} = \sum{}_{\mathrm{i=1}}^{\mathrm{n}}\,\frac{(\Delta\,\mathrm{TVDP_i})}{\Delta\,\mathrm{T}} \quad \text{for i} = 1,2,3\ldots..\mathrm{n} \tag{8}$$

$$\overline{\mathrm{IRS}} = \sum{}_{\mathrm{i=1}}^{\mathrm{n}}\left(\frac{\Delta\,\mathrm{TVRP_i}}{\Delta\,\mathrm{T}}\right) \quad \text{for i} = 1,2,3\ldots..\mathrm{n} \tag{9}$$

$$\mathrm{DDD_{pre}} = \frac{\sum{}_{\mathrm{i=1}}^{\mathrm{n}}\Delta\,\mathrm{TVDP_i}}{\mathrm{Opt.IDS}} \quad \text{for i} = 1,2,3\ldots..\mathrm{n} \tag{10}$$

$$\mathrm{DRD_{pre}} = \frac{\sum{}_{\mathrm{i=1}}^{\mathrm{n}}\Delta\,\mathrm{TVRP_i}}{\mathrm{Opt.IRS}} \quad \text{for i} = 1,2,3\ldots..\mathrm{n} \tag{11}$$

Optimal IDS and IRS are estimated using the cross-validation method. Please refer to Wu et al. (2018) for a complete description of the cross-validation method. The average relative error percentage is used to determine optimal IDS and IRS. The supplementary data shows a well-wise average relative error percentage (Table S1). Using the optimal IDS and IRS, we back-predicted development (DDD_{pre}) and recovery (DRD_{pre}) durations using Eqs. 10 and 11 for every groundwater drought episode of each well in the study area. We estimated the R^2 coefficient between observed DDD (or DRD) and predicted DDD_{pre} (or DRD_{pre}). The R^2 value is calculated as the square of the linear Pearson correlation coefficient between observed DDD (or DRD) and predicted DDD_{pre} (or DRD_{pre}). We also analyzed the spatial variability of total groundwater duration and severity in the study regions. Total Groundwater Duration (TGD) equals the sum of DDD and DRD. In contrast, severity is calculated as a cumulative sum of SGI values during the TGD. Maximum autocorrelation lag in the SGI time series significantly influences the duration of groundwater drought (Bloomfield et al. 2015). Hence, in this study, we estimated pairwise Pearson correlation coefficients between DDD, DRD, and TGD and the maximum autocorrelation lag of the SGI time series for each well. "*Plot_acf*" is a Python function used to find the autocorrelation of the SGI time series with a maximum lag of 60 months at a significance level of 5% (α=0.05). The maximum autocorrelation duration of the SGI time series is selected based on the maximum lag at which the autocorrelation factor (ACF) is more than a 95% confidence interval (Figure S2).

3.4 Groundwater drought characterization and frequency analysis

We characterized groundwater droughts into four categories: mild, moderate, severe, and extreme, based on the Standardized Groundwater Drought Index (SGI) values. The definition of drought onset and termination of various drought categories is the same as in earlier sections. Mild (MD), moderate (MoD), severe (SD), and extreme (ED)

groundwater drought conditions are observed when the SGI values are in the range (−0.5, 0), (−1.0, −0.5], (−1.5, −1.0], and (-∞, −1.5] respectively. We extracted the duration of various drought conditions for each well in the study area's three regions (NCA, SCA, RAY). We extracted the district-wise median duration of groundwater droughts under the categories for better comparison and analysis. We also performed a frequency analysis of various groundwater drought conditions using the Drought Frequency Index (DFI) (Eslamian and Eslamian. 2017). This method assigns weights from 1 to 4 for mild, moderate, severe, and extreme groundwater droughts. The Drought Frequency Index (DFI) is calculated as follows:

$$\mathrm{DFI} = \frac{\mathrm{MD_w} \times \mathrm{MD_n} + \mathrm{MoD_w} \times \mathrm{MoD_n} + \mathrm{SD_w} \times \mathrm{SD_n} + \mathrm{ED_w} \times \mathrm{ED_n}}{4\mathrm{D_l}} \quad (12)$$

Where *w*, *n* are the weights and duration of drought events under mild, moderate, severe, and extreme groundwater droughts. D_l is the total length of the groundwater time series. We also estimated the district-wide variation of extreme groundwater drought (1-month SGI values ≤ −1.5) durations in the study area.

4 Results and discussion

This study explores the variation of groundwater drought internal propagation dynamics and different drought characteristics at the district level in Andhra Pradesh state. For better comparison and analysis, districts are grouped into three regions: (1) Rayalaseema (RAY) (Chittoor-CHI, Cuddapah-CUD, Kurnool-KUR, and Anantapur-ATP), (2) North Coastal Andhra (NCA) (Visakhapatnam-VIS, Vizianagaram-VZM, Srikakulam-SRI, East Godavari-EG, and West Godavari-WG), and (3) South Coastal Andhra (SCA) (Guntur-GUN, Krishna-KRI, Prakasam-PRA, and Nellore-NEL) (Groundwater yearbook 2020-21 Andhra Pradesh, CGWB). We collected monthly groundwater and rainfall data from June 2004 to May 2021 from the India-WRIS portal, and the wells having more than 95% of the data (420 wells) were selected for analysis. We used a modified Mann-Kendall test to detect the presence of a significant trend in the selected groundwater wells. From the trend analysis, 114 wells out of 420 have no significant trend. We followed the framework developed by Wu et al. (2018) to estimate Groundwater Drought Instantaneous Propagation Speed (GDIPS) during the development and recovery stages for 114 wells distributed in the three regions mentioned above. We characterized groundwater droughts into four categories: mild, moderate, severe, and extreme, based on the range of SGI values. The Drought Frequency Index (DFI) is calculated as per Eq. 12, and different districts are classified under mild, moderate, severe, and extreme groundwater droughts. We computed district-wise median rainfall for the entire study period to analyze the correlation between monthly median rainfall and groundwater drought characteristics. The Pearson correlation coefficient is used to find the strength of linear correlation between TGD, DDD, DRD, and the maximum autocorrelation lag of the groundwater time series.

4.1 Groundwater drought development and recovery durations

Initially, we extracted groundwater drought development and recovery durations of all drought episodes of each well in three regions of the study area. Significant spatial variability exists in groundwater drought development and recovery phases along the regions of the study area (Fig. 4a and b). Cuddapah, Nellore, and East Godavari districts of RAY, SCA, and NCA regions have high DDD variation, respectively. Anantapur, Prakasam, and Srikakulam districts of RAY, SCA, and NCA regions have high median DDDs of 45.5, 43, and 54 months, respectively (Fig. 4c). The spatial variation of median DDD in all districts in three regions indicates that groundwater drought development durations are slightly higher in the NCA region compared to RAY and SCA, except in West Godavari and Vizianagaram districts (Fig. 4a). The spatial variation of median DRDs in all the districts in the three regions is opposite to the spatial variation of median DDDs (Fig. 4b). Groundwater monitoring wells in all the districts in the RAY region have a more or less similar variation of DRDs. Monitoring wells in the Prakasam district in the SCA region have significantly varying DRDs. In the NCA region, wells in the West Godavari district have a high variation in DRD. Wells in Guntur and Srikakulam districts of the SCA and NCA regions have DRDs concentrated around the median DRDs of 50.5 and 32 months, respectively (Fig. 4d). Kurnool, Guntur, and West Godavari districts of RAY, SCA, and NCA regions have high median DRDs of 50.5, 50, and 47.5 months, respectively. When comparing the overall DDD and DRD variation in the three regions, NCA has high DDD and low DRD. This shows that groundwater droughts in the NCA region take more time to reach the peak stage, and interestingly, they recover from the drought phase quickly. It is also interesting to notice that the drought development phase has a very slight variation in the three regions, whereas a significant difference is observed in the drought recovery phase. High spatial precipitation variation in traditionally drought-prone districts, Cuddapah, Chittoor, Kurnool, and Anantapur of the RAY region (Groundwater yearbook 2020-21 Andhra Pradesh, CGWB), can be attributed to the fast-developing and slow-recovering groundwater droughts in the Rayalaseema region.

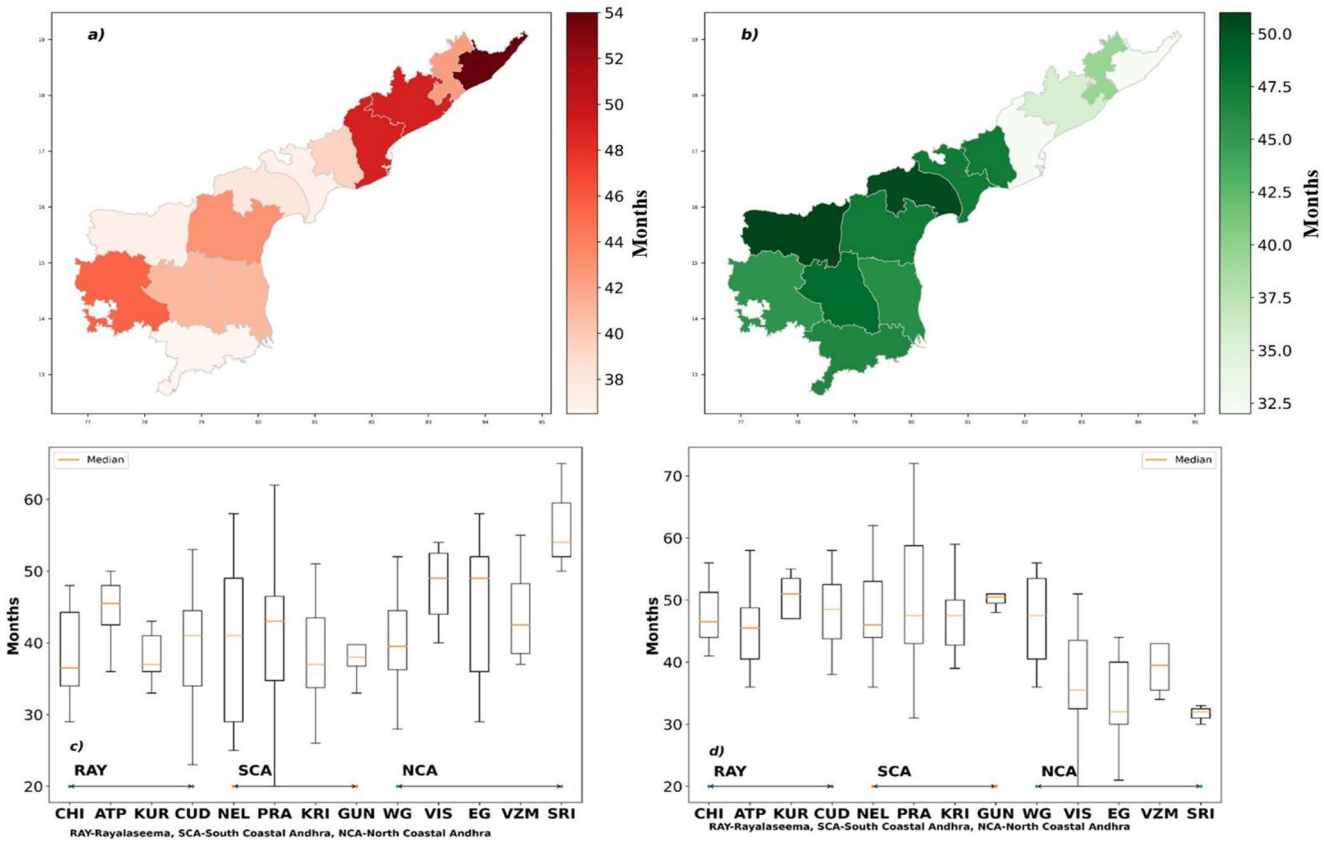

Fig. 4 (**a**) District-wise median Drought Development Duration (DDD) in the study area estimated through the GDIPS framework. (**b**) District-wise median Drought Recovery Duration (DRD) in the study area is estimated through the GDIPS framework. The lower panel shows the district-wide variation of drought development and recovery duration as box and whisker plots (**c**) and **d**), showcasing minimum, 25th, 50th, 75th percentiles, and maximum DDD and DRD in the range

The spatial contrasts in drought development (DDD) and recovery (DRD) durations across regions may reflect the combined influence of climate variability, aquifer hydrogeology, and groundwater abstraction patterns. Specifically, Rayalaseema is dominated by fractured hard-rock aquifers with low storage and limited transmissivity, whereas the North Coastal Andhra contains weathered sediments and coastal alluvial aquifers. A mix of alluvial and sedimentary aquifers occupies South Coastal Andhra (CGWB 2025). The presence of consolidated formations in the Rayalaseema region may be attributed to the rapid groundwater decline during the non-monsoon season and slow recovery during the monsoon season. In contrast, the weathered sedimentary and coastal alluvial aquifers in North Coastal Andhra allow deeper infiltration and recharge, which may result in slower drought onset and faster recovery. South Coastal Andhra shows intermediate behavior, where the Krishna-Godavari canal systems support groundwater recharge, while bore-well-intensive crop cultivation accelerates groundwater declines during non-monsoon seasons (CGWB 2025). This highlights that the spatial heterogeneity in DDD and DRD is shaped by climate–aquifer–land use interactions, not climate alone.

4.2 Optimal instantaneous development and recovery speeds

Next, we estimated well-wise optimum Instantaneous Development and Recovery Speeds (IDS and IRS). IDS and IRS provide essential information on how fast the drought episode develops and recovers to the normal situation (Wu et al. 2018; Bhardwaj et al. 2020). A significant spatial variation is observed in the district-wise median Optimal IDS and IRS values in RAY, SCA, and NCA regions (Fig. 5a and b). In the NCA and RAY regions, East Godavari and Chittoor districts observed high median optimal IDS and IRS equal to −0.28, −0.205, and 0.16,0.14, respectively. Krishna district of the SCA region observed high median optimal IDS and IRS values equal to −0.245 and 0.135, respectively (Fig. 5c and d). A high variation of optimal IDS and IRS values is observed within the districts of three regions, especially wells in Chittoor, Nellore, Vizianagaram, and

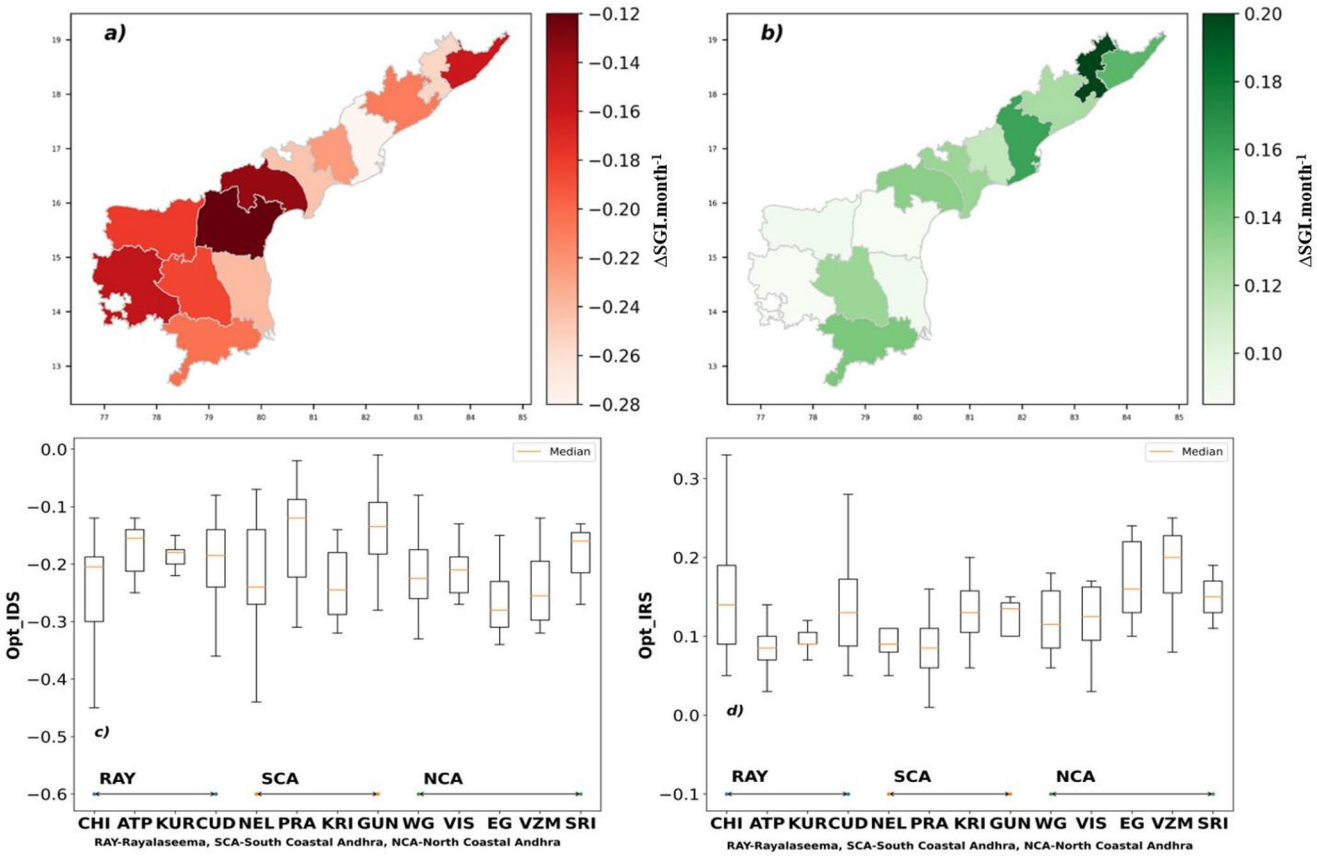

Fig. 5 (**a**) District-wise median optimal IDS in the study area estimated through the GDIPS framework. (**b**) District-wise median optimal IRS in the study area is estimated through the GDIPS framework. The lower panel shows the district-wide variation of optimal IDS and IRS as box and whisker plots (**c**) and **d**), showcasing minimum, 25th, 50th, 75th percentiles, and maximum DDD and DRD in the range

East Godavari districts of RAY, SCA, and NCA regions, observed high variation in optimal IDS and IRS values (Fig. 5c and d). The phenomenon of the same districts having high median optimal IDS and IRS values aligns with the results from (Bhardwaj et al. 2020), even though there is a slight variation in the SCA region. Anantapur and Prakasam districts of the RAY and SCA regions ascertained low median optimal IDS and IRS values. In contrast, the NCA region's Srikakulam and East Godavari districts observed low median optimal IDS and IRS values. It is interesting to notice that the NCA region has both high median optimal IDS and IRS values compared to the SCA and RAY regions, indicating rapid development and recovery of groundwater drought conditions in the NCA region. High abstract rates during non-monsoon and high rainfall during monsoon seasons could be the reason for quick groundwater drought responses during the development and recovery phases in the North coastal Andhra region. It is worth noting that optimal IDS values in all three regions are more than optimal IRS values, indicating rapid propagation of groundwater drought development conditions compared to drought recovery, which was consistent with findings from other studies (Bloomfield et al. 2015; Bloomfield and Marchant 2013; Van Loon, 2015). This pattern can be attributed to the characteristics of aquifer systems where pumping exceeds natural recharge patterns. The faster recovery observed in parts of NCA can be attributed to higher monsoon rainfall and greater aquifer storage, reducing the time required for the water table to return to normal. Conversely, the slow recovery in the RAY region may be due to low storage of hard-rock aquifers. The results also highlight that groundwater reserves are more vulnerable to the drought development phase in the study region. Over-exploitation of groundwater resources during the drought development stage can worsen the drought recovery phase and even lead to the irreversible loss of groundwater storage (Ojha et al. 2018). Using the optimal IDS and IRS values, we back predicted the DDD and DRDs for each well in all the districts (Eqs. 7 and 8). The square of the Pearson correlation coefficient is calculated to find the coefficient of determination (R^2) between observed DDD (DDD_{pre}) and DRD (DRD_{pre}). A high variation in R^2 values is observed in all the RAY, SCA, and NCA districts. However, all the districts except Vizianagaram have median R^2 values greater than

0.6 between observed and predicted DDDs. Regarding the R^2 values between observed and predicted DRDs, all districts except Visakhapatnam and Srikakulam districts have median R^2 values greater than 0.6 (Fig. 6(a, b). This highlights that optimal IDS and IRS values can predict drought development and recovery durations. IDS and IRS concepts can be effectively applied to understand the internal propagation of groundwater drought conditions.

4.3 Groundwater drought characteristics

We also estimated different groundwater characteristics like maximum and mean extreme groundwater drought durations (SGI ≤ −1.5), total groundwater drought duration equal to the sum of DDD and DRDs, and severity of groundwater droughts in all the districts of RAY, SCA, and NCA regions. High spatial variation of maximum and mean extreme groundwater durations is observed across all the districts (Fig. 7a and b). Kurnool, Prakasam, and West Godavari districts of RAY, SCA, and NCA regions have high variation in the maximum extreme durations. Cuddapah, Prakasam, and Visakhapatnam districts of the above-mentioned regions have high variation in the mean extreme groundwater drought durations (Fig. 7c and d). RAY, SCA regions have prolonged extreme groundwater droughts compared to the NCA region. This may be attributed to the high distribution of semi-critical, critical, and over-exploited groundwater wells (as per March 2017 estimates) in the Rayalaseema and South Coastal regions (Groundwater yearbook 2020-21 Andhra Pradesh, CGWB). Cuddapah, Guntur, and Visakhapatnam districts observed higher medians of maximum and mean extreme drought durations equal to 9, 7, and 7.5, and 9,5.5, and 5.5 months, respectively (Fig. 7c and d). Overall, groundwater wells in the RAY region are more susceptible to long-duration extreme groundwater droughts than in the SCA and NCA regions, consistent with CGWB (2025).

Further, we calculated the district-wise median rainfall from the monthly data obtained from the India-WRIS portal (https://indiawris.gov.in/wris/) for the study period. The idea behind estimating the median rainfall is to eliminate the influence of high rainfall and low rainfall regions in the analysis. Median rainfall is used to find the correlation between the rainfall and groundwater drought parameters like TGD, Severity, DDD, and DRDs. High spatial variation of precipitation is observed in all three regions. RAY and SCA regions have less median rainfall than NCA regions, and the RAY region is the most rainfall-stressed (Fig. 8a). As mentioned in the methodology section, groundwater drought conditions are categorized into four categories (mild, moderate, severe, and extreme). We calculated the number of mild, moderate, severe, and extreme groundwater droughts for each well in all the districts for the entire study period by aligning with the definition of drought occurrence and recovery mentioned in the methodology section. For better comparison, we calculated the median number of drought months in all the categories of droughts defined. In the RAY region, Anantapur district is found to have median drought durations of 23,22, and 24 months for mild, moderate, and severe groundwater droughts; this is complemented by the least median rainfall equal to 33.8 mm in Anantapur district (Fig. 8a). It is also interesting to find that all the districts in the RAY region have a median extreme groundwater drought duration of 12 months. Prakasam district in the SCA region, with a median monthly rainfall equal to 45.5 mm (minimum among the other districts in SCA), observed prolonged

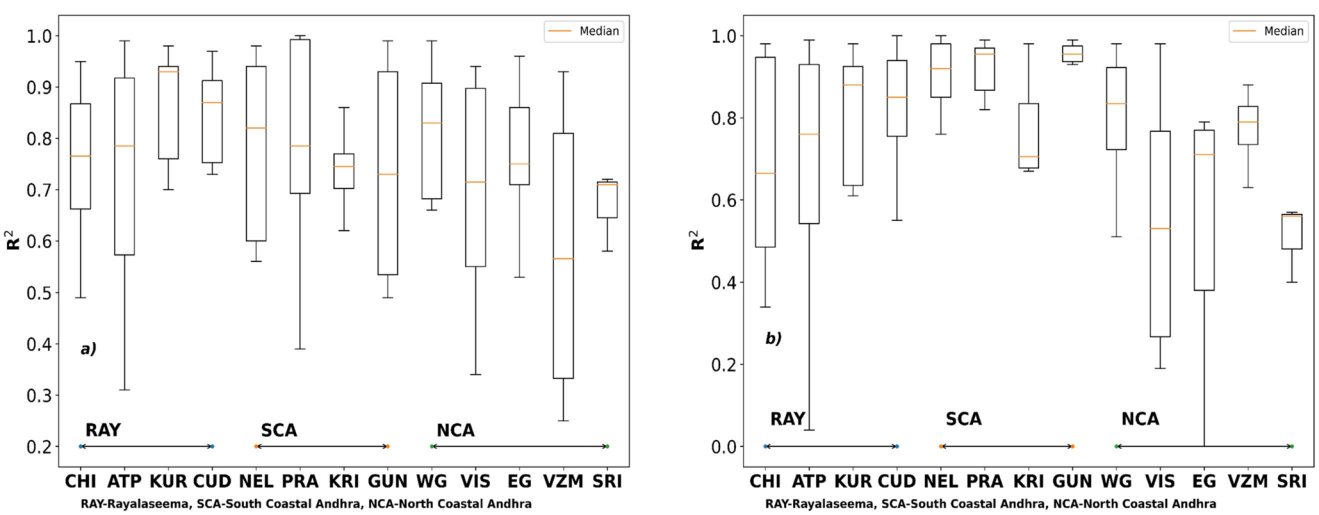

Fig. 6 (**a**) District-wide variation of R^2 values between observed DDD and DDD_{pre}, (**b**) District-wide variation of R^2 values between observed DRD and DRD_{pre} as box and whisker plots (**c**) and **d**), showcasing minimum, 25th, 50th, 75th percentiles, and maximum DDD and DRD in the range

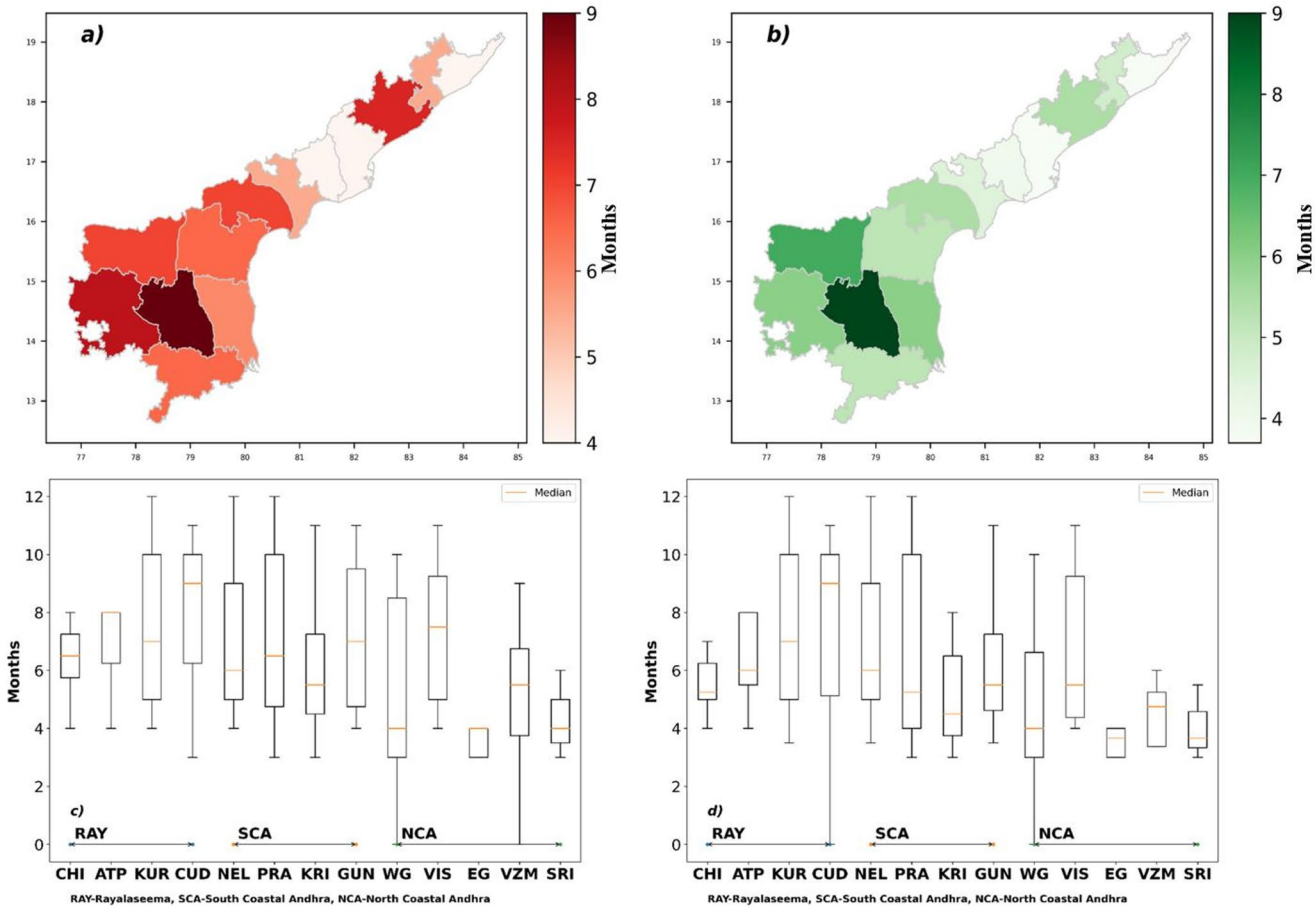

Fig. 7 (**a**) District-wide median of maximum extreme groundwater drought durations in the study area. (**b**) District-wide median of mean extreme groundwater drought durations in the study area. The lower panel shows the district-wide variation of maximum extreme and mean groundwater drought as box and whisker plots (**c**) and **d**), showcasing minimum, 25th, 50th, 75th percentiles, and maximum DDD and DRD in the range

mild, moderate, severe, and extreme groundwater droughts with median durations equal to 28, 22, 24, and 12 months, respectively. Guntur district in the SCA region, with a median rainfall of 49.6 mm, observed a maximum median drought duration of moderate, severe, and extreme drought and a minimum of median mild drought categories compared to all districts in the three regions. Interestingly, West Godavari and East Godavari districts in the NCA region, with median rainfall values equal to 48.05 and 61.01 mm, observed median mild groundwater drought durations equal to 15.5 and 26 months, respectively. Visakhapatnam district, with a maximum median rainfall of 69.4 mm, had high proportions of mild, moderate, severe, and extreme groundwater durations of 20,17,23.0, and 12 months (Fig. 8a and b). The dependence of rainfall on the occurrence of various groundwater drought categories is observed in RAY and SCA regions, with less rainfall and prolonged different groundwater drought conditions; however, in the NCA, the dependence between rainfall and groundwater drought categories is unclear; the reason for this may be attributed to the dynamic usage of groundwater resources and high rainfall amounts in the NCA region.

We also calculated Total Groundwater Drought Durations (TGDs) and Severity for each well in the study region's districts. DDD and DRDs are subsets of TGD, and high spatial variation of TGDs is observed in RAY, SCA, and NCA regions. Overall, RAY and SCA regions have higher TGDs than the NCA regions. The same severity levels of groundwater droughts are observed in RAY and SCA regions; this can be attributed to more severe, moderate, and extreme groundwater droughts than the NCA region (Fig. 8c and d). We also performed a correlation analysis between district-wise median rainfall and groundwater drought characteristics (TGD, Severity, DDD, and DRD). Table 1 provides the Spearman correlation coefficients between monthly median rainfall and groundwater drought characteristics. TGD and median rainfall are highly correlated with a Spearman Rank Correlation Coefficient (SCC) equal to −0.66 (negatively correlated), i.e., a decrease in rainfall increases total groundwater duration. Severity and rainfall are significantly

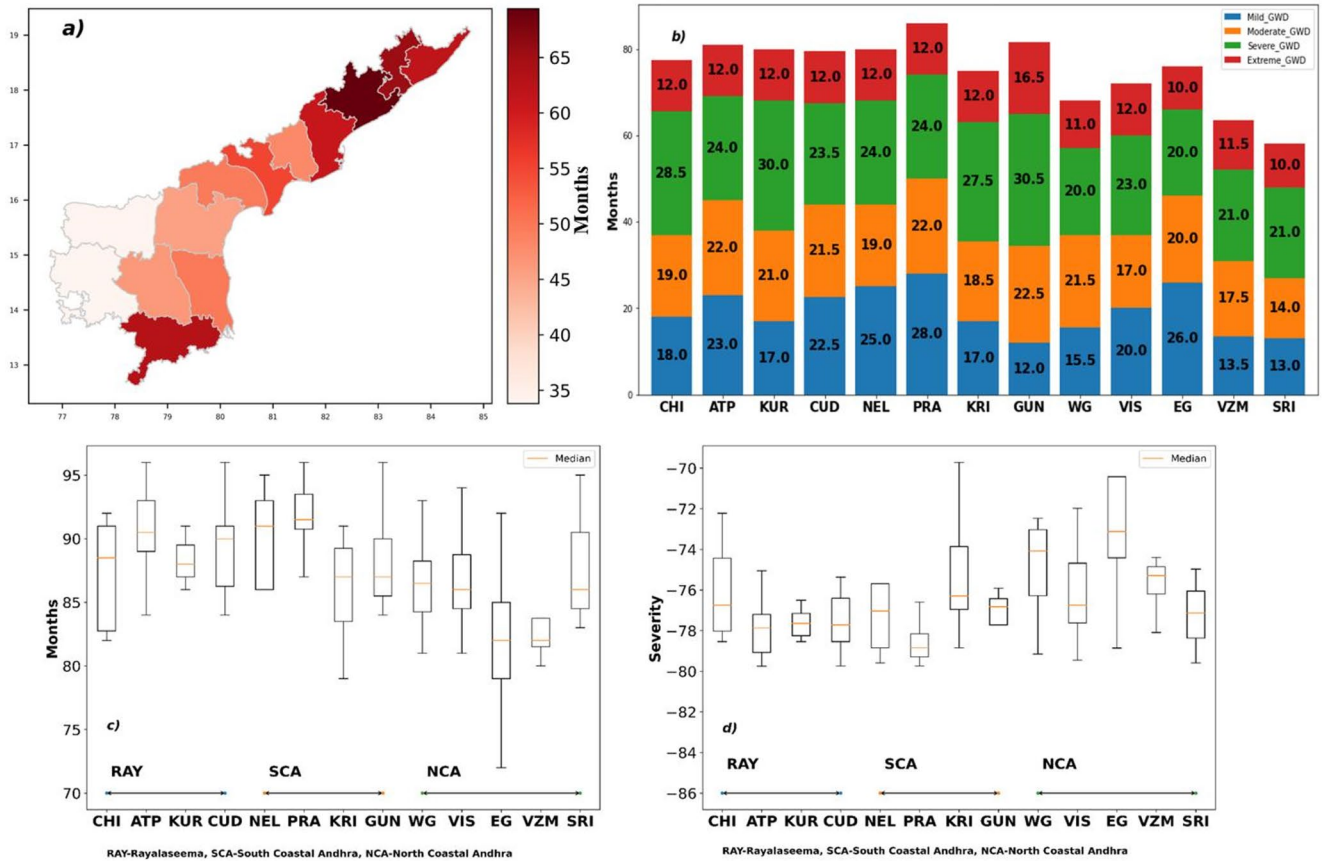

Fig. 8 (**a**) District-wise monthly median rainfall in the study area. (**b**) District-wide variation of the median of mild, moderate, severe, and extreme groundwater drought durations in the study area. The lower panel shows the district-wide variation of Total Groundwater Drought Duration (TGD), and Severity as box and whisker plots (**c**) and **d**), showcasing minimum, 25th, 50th, 75th percentiles, and maximum DDD and DRD in the range

Table 1 Correlation coefficient between median values of (TDD, Severity, DDD, and DRD) and rainfall for Andhra Pradesh

Variable1	Variable2	SCC*	*P*-value*
TGD	Rainfall	−0.66	0.013
Severity		0.63	0.019
DDD		0.19	0.521
DRD		−0.62	0.025

*SCC: Spearman Correlation Coefficient, *P-value at 5% significance

correlated with SCC = 0.64; here, this relation should be carefully interpreted because the increase in median rainfall results in a decrease in severity, but the severity is denoted in negative values; hence, more rainfall results in less severity by moving towards a positive direction. No significant correlation is observed between DDD and rainfall, with SCC = 0.19; this may be due to various other factors affecting the DDD and the rainfall. Interestingly, DRD is significantly and indirectly correlated to the median rainfall with SCC = −0.62, i.e., high monthly median rainfall results in low DRDs. This highlights that rainfall distribution is crucial in groundwater systems recovering from drought.

The concentration of long-duration severe and extreme droughts in RAY and SCA can be linked to both climatic aridity and high groundwater abstraction for irrigation. Semi-arid regions like RAY experience low and high variable rainfall with high evapotranspiration, resulting in limited natural recharge, as Shaikh et al. (2023) reported similar findings in the spatial and temporal analysis of drought in southern India river basins. Meanwhile, districts such as Prakasam and Guntur have high cropping intensities and widespread cultivation of water-intensive crops, which produce persistent abstraction pressures that may result in more extreme groundwater drought characteristics. The same aspect has been discussed by Jain et al. (2021) while studying the effect of cropping intensity on groundwater depletion. Canal networks and alluvial aquifers in coastal regions increase drought recovery resilience, which can be observed with mild to moderate groundwater drought conditions.

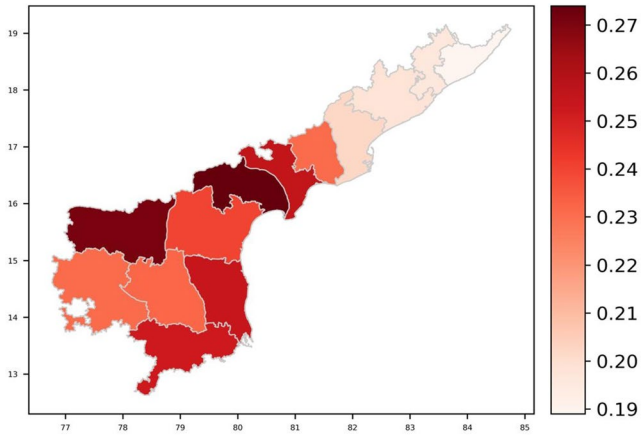

Fig. 9 District-wise median Drought Frequency Index (DFI)

Table 2 Weighted classification scheme of drought frequency index (DFI) (Eslamian and Eslamian 2017)

DFI range	Drought frequency class
0 < DFI ≤ 0.20	Low
0.20 < DFI ≤ 0.40	Moderate
0.40 < DFI ≤ 0.60	High
0.60 < DFI ≤ 1.00	Very High

4.4 Groundwater drought frequency analysis

This study analyzed groundwater drought frequency using the Drought Frequency Index (DFI) (Eslamian and Eslamian 2017). DFI is calculated for each well in all the districts of the study region (RAY, SCA, NCA). For better comparison, we calculated the district-wise median DFI for RAY, SCA, and NCA regions (Fig. 9). Table 2 summarizes different drought frequency classes based on DFI values; interestingly, all the districts in the RAY and SCA regions have DFI values ranging between 0.20 and 0.40. According to Table 2, all the districts in the RAY and SCA regions come under the moderate drought frequency class. In contrast, in the NCA region, except for West Godavari, all the districts have DFI values in the range [0, 0.20] and are in the mild drought frequency class. Kurnool, Guntur, and West Godavari districts in RAY, SCA, and NCA regions have maximum median DFI values equal to 0.27,0.27, and 0.23. However, it should be noted that the mentioned median of DFI values of different wells in all the regions gives an approximate representation of district-wise groundwater drought frequency. This identification of mild and moderate groundwater drought frequency classes agrees with many studies that identified prolonged and less severe groundwater droughts (Bloomfield et al. 2015; Bloomfield and Marchant 2013; Guo et al. 2021; Kumari et al. 2023).

4.5 Correlation between maximum autocorrelation lag and groundwater drought durations

The autocorrelation of the SGI time series is expected to be directly interlinked with the duration of the groundwater drought. This behavior is typical of unconfined fractured aquifers, where storage is primarily controlled by the thickness of the weathered zone and fractured connectivity, resulting in slow recession and extended recovery durations. The high autocorrelation increases the time required for the groundwater drought conditions to reach normal or non-drought conditions (Bloomfield et al. 2015). In this study, we calculate the pairwise Pearson correlation coefficient (PCC) between well-wise DDD (or DRD or TGD) and the maximum autocorrelation lag of the groundwater time series. We are more interested in the positive correlation coefficients because positive correlation implies a direct linear relationship between the variables. High variation is observed in the linear correlation between the maximum autocorrelation lag of the groundwater time series and various groundwater drought durations (Fig. 10). Anantapur and Cuddapah districts of the RAY region observed

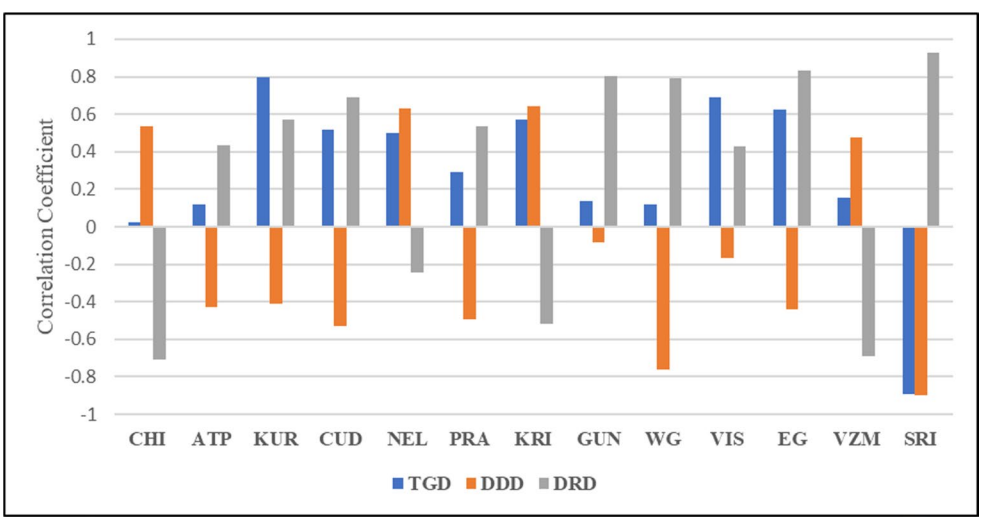

Fig. 10 District-wise Pearson correlation coefficients (PCC) between (DDD, DRD, TGD) and the maximum autocorrelation lag

maximum positive PCC (0.43,0.69) between DRD and maximum autocorrelation lag. In contrast, DDD and TGD of Chittoor and Kurnool districts of the same RAY region are highly correlated to maximum autocorrelation lag (PCC = 0.54, 0.79), respectively. In Nellore and Krishna, Prakasam, and Guntur districts in SCA regions, DDD and DRD are positively correlated to the maximum autocorrelation lag of the groundwater time series with PCC equal to 0.63,0.64,0.54,0.80, respectively. In the NCA region, West Godavari, East Godavari, and Srikakulam districts observed DRD has high positive PCC equal to 0.79,0.83, and 0.92, respectively, whereas in Visakhapatnam and Vizianagaram districts DDD and TGD have high positive PCC equal to 0.69 and 0.74, respectively. In the majority of districts in three regions, we can see a significant positive correlation (PCC > 0.6) between various groundwater drought durations and the maximum autocorrelation lag of groundwater, with some exceptions at Anantapur, Chittoor, Prakasam, and Vizianagaram districts. In the present study, we did not consider the effects of the occurrence and deficit of precipitation during drought and recovery phases, which alter groundwater drought's development and recovery stages due to the availability and non-availability of recharge to groundwater.

The positive correlation between maximum autocorrelation lag and groundwater drought duration indicates that aquifers with greater hydrological memory experience more prolonged drought propagation (Bloomfield et al. 2015). Hard-rock and fractured aquifers in Rayalaseema exhibit short memory and slow recovery, consistent with limited storage and restricted lateral recharge connectivity of consolidated aquifers as mentioned by Taylor et al. (2013) and Imane et al. (2025). In contrast, alluvial and deltaic aquifers in the coastal plains show longer memory effects, enabling faster drought termination and being less susceptible to groundwater droughts, which is consistent with the findings of Schuler et al. (2022).

5 Limitations and future work

The present study did not consider the effects of the occurrence and deficit of precipitation during the drought development and recovery phases, which alter the development and recovery stages of groundwater droughts due to the availability and non-availability of recharge to groundwater. Incorporating the above-mentioned aspect could improve understanding of groundwater drought internal propagation mechanisms at regional scales in the study area. Moreover, in this study, we did not incorporate the influence of abstraction rates and LULC changes on the internal propagation of groundwater droughts. Integrating LULC maps and abstraction rates could significantly improve understanding of complex groundwater drought propagation mechanisms under anthropogenic activities, and is considered to be of future scope.

6 Conclusions

This study analyzed various groundwater characteristics across multiple districts in three regions of Andhra Pradesh state. The Groundwater Drought Instantaneous Propagation Speed (GDIPS) framework is developed to understand better the internal propagation mechanisms of groundwater droughts across different regions of Andhra Pradesh. The standardized groundwater index (SGI) is used to identify groundwater drought events and their respective drought characteristics. We also employed correlation analysis between rainfall and groundwater drought durations to understand the role of rainfall in groundwater drought dynamics. Groundwater drought frequency analysis is performed to identify the occurrence of multiple categories of groundwater droughts in the study area. We also performed correlation analyses between the groundwater drought durations and the groundwater timeseries auto-correlation function.

From the significant findings of the present study, the following conclusions can be framed:

1. Groundwater drought development and recovery durations exhibit pronounced spatial variability across Andhra Pradesh. Districts in North Coastal Andhra experience longer drought development but faster recovery, while Rayalaseema and South Coastal Andhra districts showed the opposite pattern, with rapid development and slower recovery.
2. The Groundwater Drought Instantaneous Propagation Speed (GDIPS) framework and optimal instantaneous speeds (IDS/IRS) effectively capture and predict the internal propagation dynamics of groundwater droughts. Strong correlations between observed and predicted drought durations validate the utility of these metrics for regional groundwater drought analysis.
3. Groundwater droughts in Andhra Pradesh generally develop more quickly than they recover, indicating that aquifer systems are more vulnerable to rapid depletion and slower replenishment, which poses challenges for sustainable management.
4. Extreme, severe, and long-duration groundwater droughts are predominantly concentrated in Rayalaseema and South Coastal Andhra, making these regions particularly susceptible to groundwater stress and prolonged water scarcity.
5. The frequency of groundwater droughts in the study area is mostly moderate to mild, with regional differences reflecting underlying hydrogeological and climatic variability.

6. Rainfall variability is a significant driver of groundwater drought propagation. Drought recovery and total groundwater drought durations strongly correlate to rainfall patterns, underscoring the importance of precipitation in groundwater system resilience.
7. A key limitation of this study is the exclusion of anthropogenic influences such as LULC changes, irrigation practices, and groundwater abstraction rates. While the analysis provides valuable insights into natural groundwater drought dynamics, future research should integrate these human-driven factors to fully understand and manage the complexity of groundwater drought propagation in Andhra Pradesh.
8. From a management perspective, groundwater drought mitigation in Andhra Pradesh requires region-specific strategies. Recommended measures include artificial recharge structures in drought-prone districts, promoting efficient irrigation technologies, and regulating unsustainable pumping practices. These measures may strengthen the resilience of groundwater resources in the study area.

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Author contributions P.C.: Conceptualization, Investigation, Methodology, Formal analysis, Writing- original draft, review, and editing. N.C: Data curation, Methodology, Formal analysis.

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Data availability The groundwater and rainfall data used in this study were obtained from the publicly available India-WRIS portal. Processed datasets can be made available upon reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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